Numerical Analysis Assignments

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1. Determine the order of the local truncation error of the following method:

$$Y_{i+1} = Y_i + hf(x_i + \frac{h}{2}, Y_i + \frac{h}{2}f(x_i, Y_i)).$$

2. Here is an implicit midpoint Runge-Kutta method (one stage):

$$Y_{i+1} = Y_i + hf(x_i + \frac{h}{2}, \frac{Y_i + Y_{i+1}}{2}).$$

   Verify its order using Taylor expansion.

3. Consider the explicit trapezoidal method, a two-stage Runge-Kutta method:

$$\begin{aligned} K_1 &= f(x_i, Y_i) \\ K_2 &= f(x_i + h, Y_i + hK_1) \\ Y_{i+1} &= Y_i + \frac{h}{2}(K_1 + K_2). \end{aligned}$$

   Stability analysis of the method is by applying it to the scalar test equation $y' = \lambda y$ where $\Re(\lambda) < 0$. Sketch or describe the stability region in the complex plane, and determine the maximum step size h for stability when $\lambda = -100 + 10i$.

4. For the two-step Adams-Bashforth method:

$$Y_{i+1} = Y_i + \frac{h}{2}[3f(x_i, Y_i) - f(x_{i-1}, Y_{i-1})]$$

expand the exact solution $y(x_{i+1})$ using Taylor series around x_i and derive the local truncation error τ_{n+1}, identifying its order.

5. Construct a three-step explicit Adams-Bashforth method. Assume the form

$$Y_{i+1} = Y_i + h\sum_{j=0}^{2} \beta_j f(x_{i-j}, Y_{i-j}).$$

Determine $\beta_0, \beta_1, \beta_2$ to achieve third-order accuracy by matching the Taylor expansions at x_{i+1}.

6. Check the zero-stability of the following three-step method

$$Y_{i+1} - 3Y_i + 3Y_{i-1} - Y_{i-2} = h[f(x_{i+1}, Y_{i+1}) - f(x_{i-2}, Y_{i-2})].$$